Total No. of Questions : 8] | Total No. of Printed Pages : 3

Paper Code : 21306

F-406

B.C.A. (Second Semester) Examination, 2023

(New Course)

Papar No. BCA-(N)-201

DIGITAL ELECTRONICS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. Perform the following conversions:
   (a) Decimal number 56 to its binary equivalent
   (b) Binary number 110110 to its decimal equivalent

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   (c) Octal number 267 in its decimal equivalent
   (d) Hexadecimal number F3 to its binary equivalent
2. Discuss the concept of Karnaugh map. Solve the following Boolean function using Karnaugh map:
   F (A, B, C, D) =Σ (0, 1, 3, 5, 7, 9, 12, 13, 14)
3. Using De Morgan's laws, simplify the following Boolean expression:
   (A+B) + (C+D)
4. What is a binary to decimal decoder in digital electronics? Explain how it can be used to convert a binary number to its decimal equivalent.
5. What is the difference between a multiplexer and a demultiplexer in digital electronics? Provide examples of each.

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6. What is a JK flip-flop? Explain how it can be used to implement both synchronous and asynchronous operations.

7. Define the term "Race around condition" and explain how it can be avoided.

8. Explain the concept of "propagation time delay" in digital circuits. How does it affect the performance of a digital system?

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